

OPERATIONAL STUDY OF CIRCULATING FLUIDIZED BED STEAM GENERATOR

The [original paper](#) contains 38 sections, with 7 passages identified by our machine learning algorithms as central to this paper.

Paper Summary

SUMMARY PASSAGE 1

Introduction

Circulating Fluidized bed combustion (CFBC) steam generator is equipment, used for generating steam by burning the fossil fuels in a furnace where finer solids are transported through the furnace with a velocity higher than the terminal velocity of the particles. The solids, leaving the furnace is separated by a solid-gas separator (cyclone) and is circulated back to the furnace through the down comer, seal loop, spiess valve and the return leg.

SUMMARY PASSAGE 2

The Thermal System

The Circulating Fluidized Bed (CFB) unit consists of the combustor, water and steam system, air and flue gas system, fuel and ash system, auxiliary systems like gas analysers, feeding system, solid handling system etc.

SUMMARY PASSAGE 3

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The CFB combustor consists of the furnace, cyclone and circulating system (down comer, seal loop, spiess valve, the return leg and the Fluidized bed heat exchanger (FBHE)). The convective section has the reheater coils. The combustion takes place in a vertical furnace in which the fluidization occurs.

SUMMARY PASSAGE 4

Operational Study

The combustion air system consists of two primary air fans, two secondary air fans, rotary piston blower and empty chambers. Primary air is used to remove the moisture in the fuel. Secondary air provides the air for combustion and fluidizing air. Induced draft fan absorbs the air and the exhaust gas is expelled out of the stack to the atmosphere.

SUMMARY PASSAGE 5

Flue Gas Analysis

Flue gas carry large amount of heat which can be utilized to improve the performance of the steam generator.

SUMMARY PASSAGE 6

Figure 2: Oxygen Concentration Vs. Steam Generator Efficiency

The heat energy is utilized for various applications which includes boilers, engines, etc. The combustion efficiency of a circulating fluidized bed steam generator varies from 98% to 99.9%. Also, it has better solid-gas mixing i.e. turbulence is provided for the combustion. Fluidized bed combustor is metallurgic ally stable.

SUMMARY PASSAGE 7

Conclusions

The operational parameters of the lignite fired circulating fluidized bed combustion steam generator were studied. It was found that particle size plays an important role in the efficiency of the steam generator. Finer particles exhibit better heat transfer characteristics than the coarser particles.